

PROJECT EVIDENCE

CemeteryLink Enrichment Agent

Operational workflow for turning documented burial data into reviewable cemetery and grave records

26 September 2026

Project summary

CemeteryLink already had broad cemetery coverage, but grave-level content was sparse and manual enrichment did not scale. I built and operationalized a Python workflow that finds documented burials, resolves or creates the cemetery, creates or enriches the grave record, and keeps an auditable SQLite state. The strongest demonstrated value is grave coverage at scale, with conservative handling of uncertain evidence.

Ireland V2	Ireland V2	UK validation	UK validation
1,051 operations	1,045 distinct graves	20 candidates	0 runtime errors
314 CREATED / 737 UPDATED	236 cemetery IDs associated	19 cemeteries / 20 graves	5 / 5 reviewed outputs accepted

Evidence sources: SQLite audit data, live CemeteryLink outputs, terminal run logs and a documented second-user test.

Project overview

The problem

CemeteryLink had broad cemetery coverage from imported location datasets, but many records still needed cleanup and most cemetery pages had little or no burial content. Adding graves manually meant repeating the same research and data-entry sequence: find a documented burial, identify the correct cemetery, check whether it already existed, create it if necessary, check for an existing grave, and then add or improve the person record.

That process was workable for individual records but not for building grave coverage at scale. The project needed a repeatable workflow that could research, match, publish and resume safely without turning generated text into an authority.

What I designed and built

I built a Python workflow that takes burial candidates from open structured sources and turns them into controlled CemeteryLink updates. The workflow combines source retrieval, cemetery matching, location research, WordPress/CemeteryLink API calls and SQLite state tracking.

The model is used for web-assisted cemetery research and concise factual descriptions when source material supports them. The burial relationship itself comes from source data; entity matching, duplicate/state checks and publication rules are handled by code rather than delegated to free-form generation.

How it works in practice

1. Retrieve people with a documented burial place from the configured open-data source.
2. Match the cemetery against CemeteryLink using normalized names, coordinates and administrative context.
3. If the cemetery is missing, research the place and create it only when there is enough location evidence.
4. Create a new grave, or enrich an existing grave when the incoming information adds useful structured data.
5. Store source identifiers, outcomes and run state in SQLite so the workflow can be inspected and continued.

Responsible AI and human boundaries

AI may	Research cemetery context and draft concise factual descriptions from supported source material.
Code / rules	Control source retrieval, matching logic, state, duplicate checks and whether the workflow proceeds.
AI must not	Invent a burial place, date, occupation, biography or historical fact when the evidence is absent or conflicting.
Human role	Set the scope and configuration, review representative outputs, investigate exceptions and decide whether to expand the workflow.

Evidence so far

Evidence	Observed result	Status
Ireland production	1,051 operations: 314 CREATED and 737 UPDATED; 1,045 distinct grave IDs; 236 cemetery IDs associated; 140 cemetery-research entries; cursor completed.	Measured
Recorded throughput	Run window 25 Sep 22:38 to 26 Sep 00:57 (about 2 h 19 min), equivalent to ~454 processed operations/hour across that recorded window.	Measured
Development iteration	Earlier broad run: 8,335 candidates; 6,769 skipped for "cemetery not found and coordinates missing". This failure mode drove the V2 cemetery-resolution step.	Measured
Second-user validation	20 UK candidates across two controlled batches; 19 cemeteries and 20 graves created; 0 runtime errors. tester manually reviewed 5 outputs and accepted all 5 without correction.	Measured + reviewed sample

What the second-user test exposed

The content review itself was straightforward, but the initial startup command and Python environment were less intuitive. The operating instructions were tightened so the worker can be launched with the virtual-environment Python directly and can load its project configuration from the local .env file.

What I learned / what comes next

The main value is grave coverage, not cemetery prose. Grave records can contain useful structured biographical data even when the cemetery page remains basic. That limitation is visible in the evidence and is not hidden behind generated descriptions.

The next evidence step is a small manual baseline on comparable records. Until that is measured, I treat the recorded throughput and tester feedback as operational evidence rather than a quantified time-saving or financial ROI claim. For expansion, I will continue monitoring cemetery matching, unsupported facts, correction rate and review effort.

Scope and limitations

- The five UK outputs reviewed by the second user are a small quality sample, not a global accuracy rate.
- A controlled manual baseline has not yet been measured, so no percentage time-saving or financial ROI is claimed.
- Cemetery descriptions remain uneven when source material is limited; grave-level enrichment is currently the stronger output.
- Production writes are automatic after the workflow's checks, so auditability, representative review and conservative matching remain important controls.

Design iteration: from failure mode to V2

Observed V1 bottleneck	Design change	V2 outcome
6,769 candidates skipped because the cemetery was not found and coordinates were missing.	Resolve or research the cemetery first; create it when evidence is sufficient; then continue to the grave.	Ireland V2 completed 1,051 operations and affected 1,045 distinct grave IDs across 236 cemetery IDs.

Evidence annex

A. Ireland production examples

These screenshots are representative production outputs. They illustrate what the workflow publishes, not a statistical quality sample.

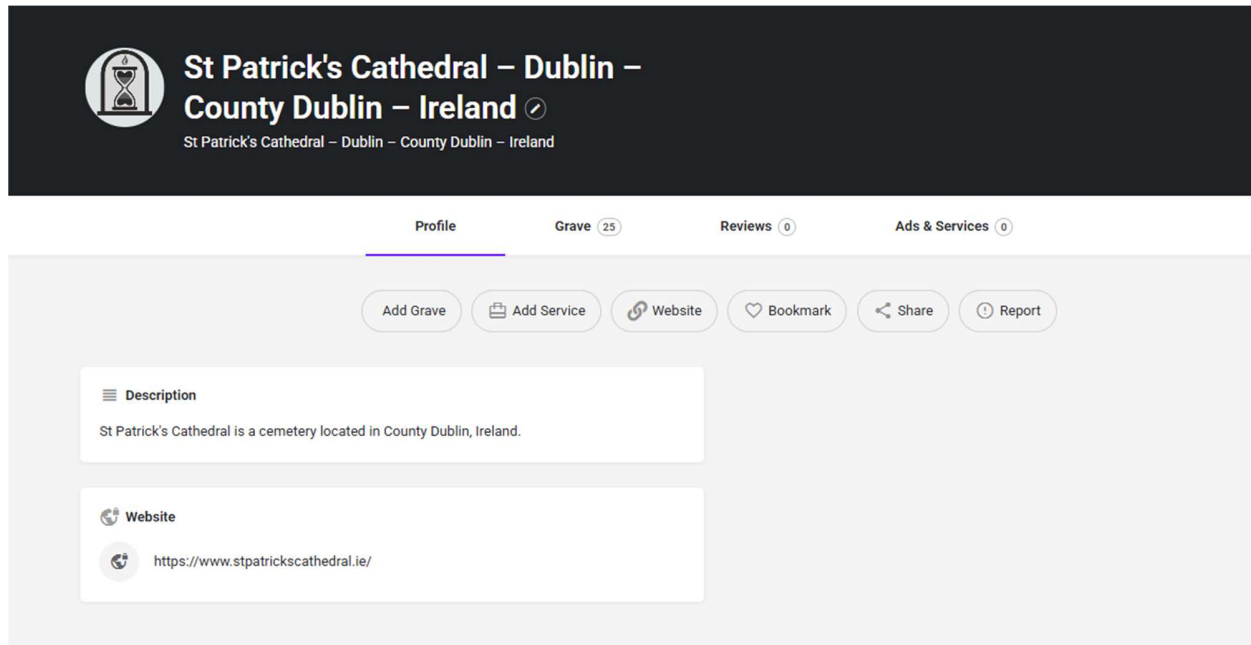


Figure 1. St Patrick's Cathedral, Dublin — an existing cemetery now showing 25 linked graves.

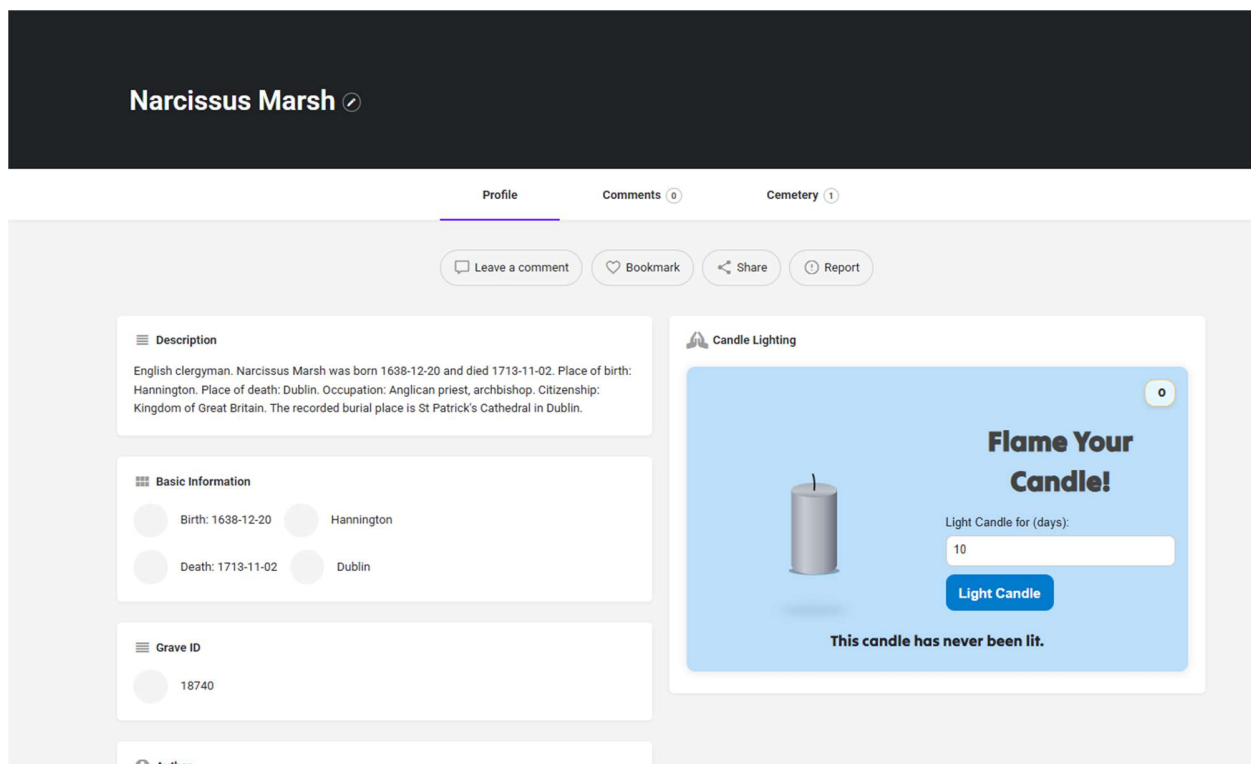


Figure 2. Narcissus Marsh — structured grave record linked to St Patrick's Cathedral.

A. Representative production outputs (continued)

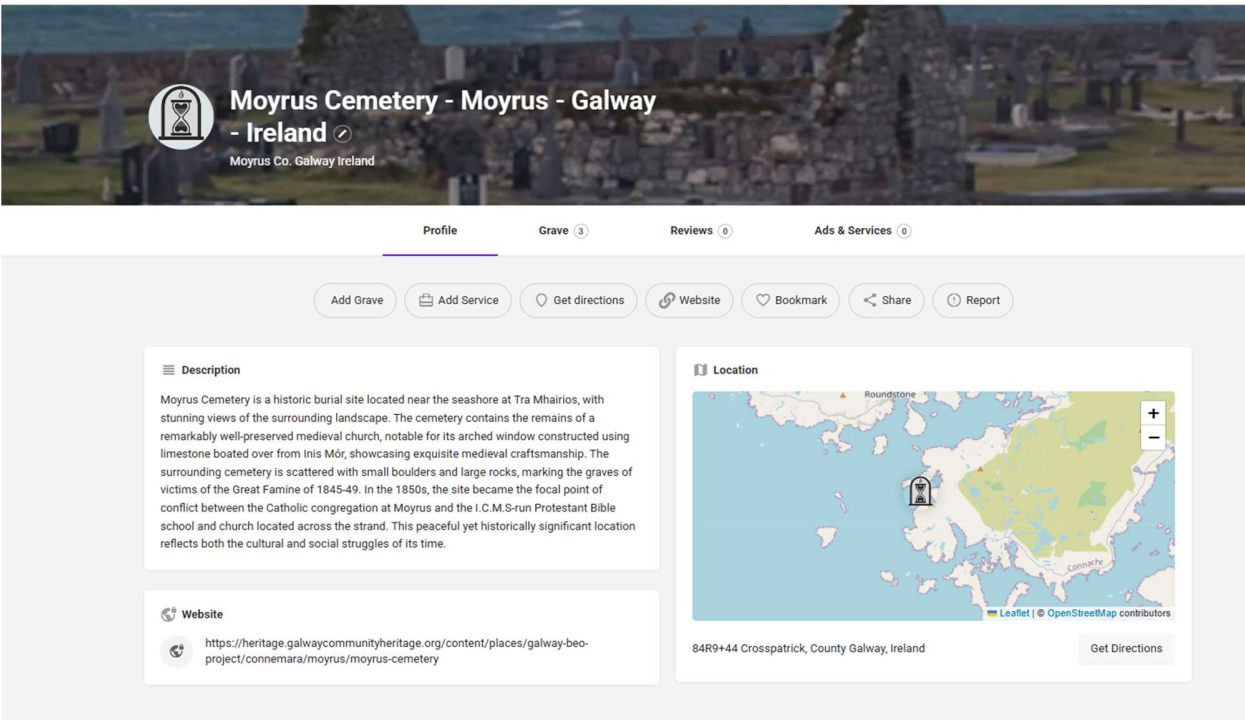


Figure 3. Moyrus Cemetery, Galway — stronger cemetery-level enrichment with researched context, location and external source.

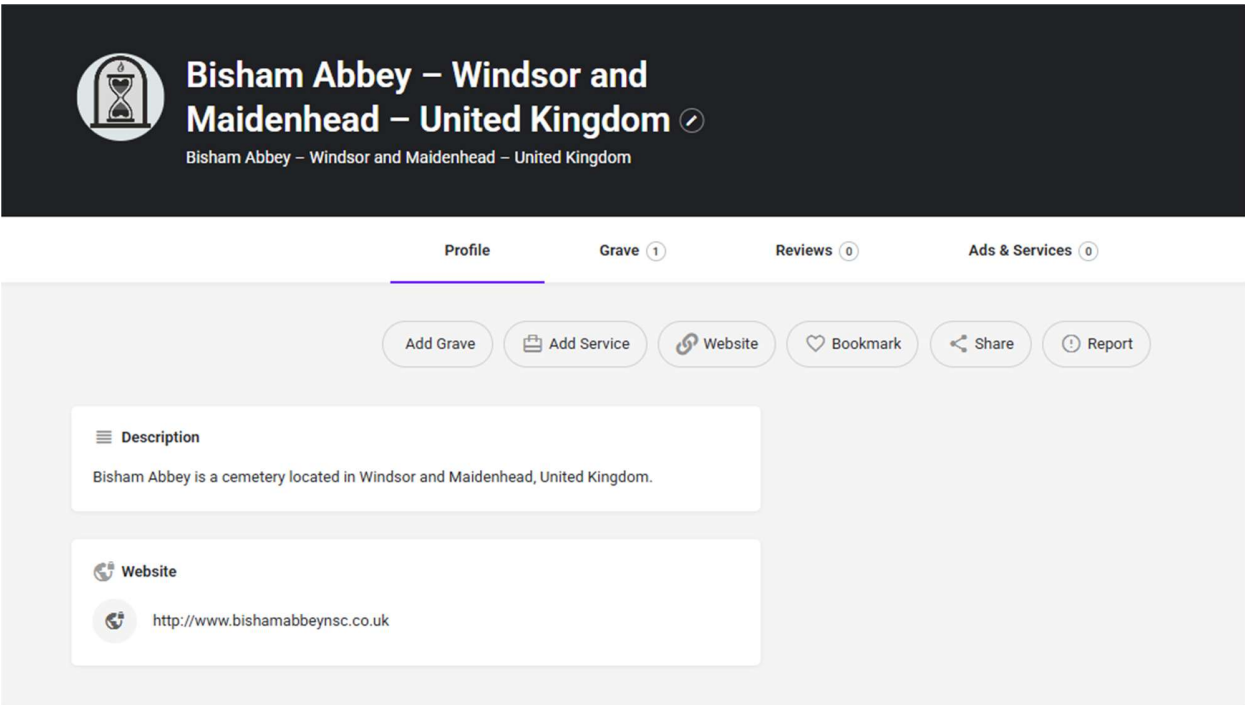


Figure 4. Bisham Abbey, United Kingdom — cemetery created during the second-user validation.

B. Second-user validation evidence

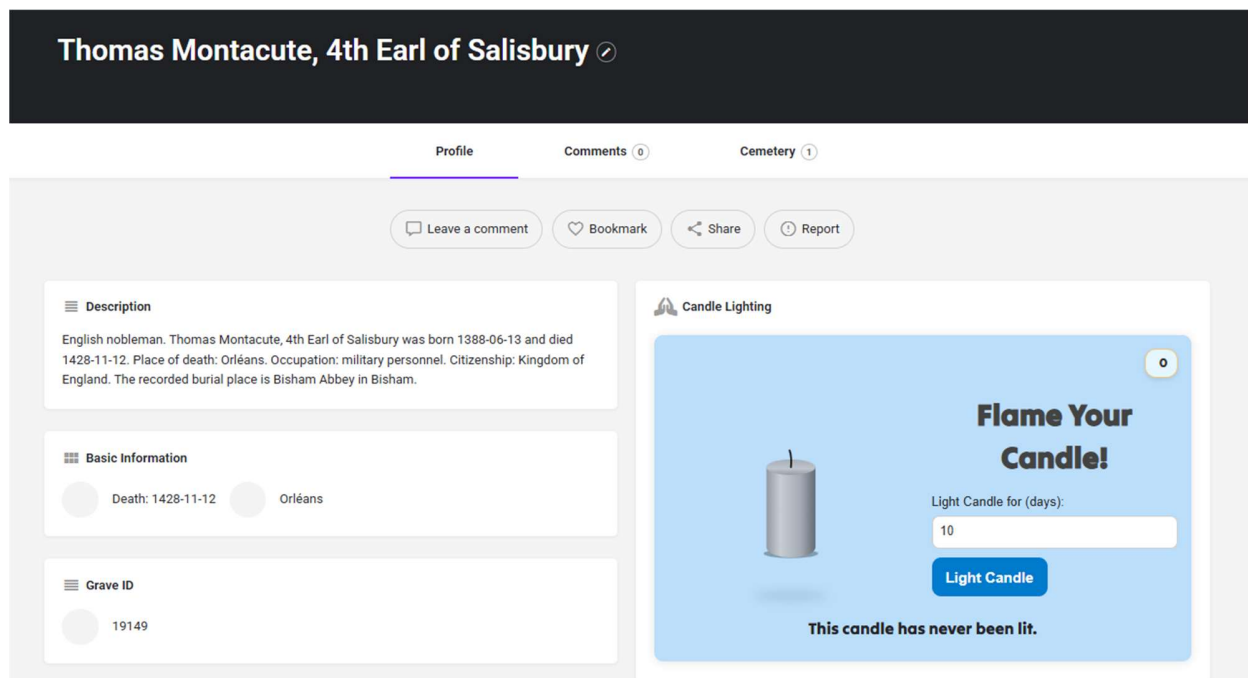


Figure 5. Thomas Montacute, 4th Earl of Salisbury — grave created and associated with Bisham Abbey during the UK validation.

```
CEMETERY CREATED: 19183
CEMETERY ENRICHED: ['description', 'location']
GRAVE CREATED: 19184
https://cemeterylink.com/other/helen-spurrell/

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PERSON: Robert Burn
CEMETERY: Old Calton Burial Ground
ADMIN: City of Edinburgh
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NEW CEMETERY: Old Calton Burial Ground
OSM SEARCH: Old Calton Burial Ground, City of Edinburgh, United Kingdom
RESEARCHING CEMETERY: Old Calton Burial Ground
CEMETERY CREATED: 19185
CEMETERY ENRICHED: ['description', 'postal_code', 'location']
GRAVE CREATED: 19186
https://cemeterylink.com/other/robert-burn/

=====
TEST LIMIT REACHED
People handled: 10
Cemeteries created: 10
Graves created: 10
Graves updated: 0
Errors: 0
=====
```

Figure 6. Second controlled UK batch — 10 people handled, 10 cemeteries created, 10 graves created, 0 runtime errors.

C. Second-user validation record

Test date: 26 September 2026 **Environment:** Preconfigured CemeteryLink development PC **Scope:** United Kingdom, 20 candidates across two controlled batches

Check	Tester response
Able to run the workflow	Yes
Most confusing part	Technical setup and the exact command used to start the worker
Help required	Initial clarification about how to launch the Python worker
Output and grave-to-cemetery association clear	Yes
Compared with manual research/data entry	Much faster (reported, not timed)
Comfortable repeating the workflow	Yes
Improvement suggested	Simpler launch instructions; richer information on some cemetery profiles

Tester comment: "After seeing the process once, the steps were straightforward and the results could be checked directly in CemeteryLink."

D. Evidence ledger

Evidence	What it supports	Status / limitation
Ireland V2 SQLite audit	Counts, distinct IDs, run dates and completed cursor	Measured
Historical SQLite state	Failure mode that drove the redesign	Measured; broader scope
Live CemeteryLink screenshots	Examples of cemetery and grave outputs	Observed examples
UK terminal logs + validation	20 candidates, 19 cemeteries, 20 graves, 0 errors; 5-result manual review	Measured + reviewed sample
README + evidence workbook	Repeatable operating instructions and extracted evidence	Reusable supporting assets

E. Supporting evidence package

- README_CemeteryLink_Enrichment_Agent.txt — operating instructions, scope and known limitations.
- CemeteryLink_SQLite_Evidence.xlsx — extracted metrics, examples and V1/V2 evidence summary.
- CemeteryLink Enrichment Agent – Production Workflow Demo.mp4 — short walkthrough showing a run and the resulting CemeteryLink records.